# An open basin of four-point limits for a Markov distribution mapping

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**Task 1. Resolve the finite metric-space convergence conjecture.** Conjecture 5 of Aldous–Cruz–Feng [1], recorded as MathDB 362490, asserts that on every finite metric space with distinct interpoint distances, almost every initial distribution has iterates under $\pi_{j,k}$ converging to a point mass or a uniform distribution on two points. Here $\pi_{j,k}(p)$ is the *stationary distribution* of the chain that samples $k$ independent points with law $p$ and moves to the $j$th closest to its current state. The asserted almost-everywhere convergence is false, already for $(j,k)=(2,2)$.

**Theorem 1.** *Let $S=\{1,2,3,4\}$ have distance matrix*

$$D=\frac{1}{10}\begin{pmatrix}0&15&13&11\\15&0&12&14\\13&12&0&16\\11&14&16&0\end{pmatrix}.$$

*There is a nonempty open subset of the three-dimensional probability simplex such that every distribution in that subset has $\pi_{2,2}$-iterates converging to $(a,a,\frac12-a,\frac12-a)$ for some $0<a<\frac12$.*

*Proof.* All six interpoint distances are distinct. They lie between 11/10 and 16/10, so every triangle inequality holds strictly. The rank matrix, with rank one assigned to the current point itself, is

$$R=\begin{pmatrix}1&4&3&2\\4&1&2&3\\3&2&1&4\\2&3&4&1\end{pmatrix}.$$

Write $F(p)=\pi_{2,2}(p)$ for interior probability vectors, using row vectors. Its transition matrix is the quadratic polynomial matrix

$$K_{xy}(p)=p_y^2+2p_y\sum_{z:R_{xz}<R_{xy}}p_z. \tag{1}$$

Indeed, either both samples equal $y$, or one equals $y$ and the other is strictly closer to $x$. This also covers the possibility that a sample equals the current state. Every entry is positive for interior $p$, so the stationary vector is unique. Moreover $F$ is analytic there: the linear equations $FK=F$ and $\sum_y F_y=1$ have a unique solution, which is locally a rational function of the entries of $K$.

First observe the entire line of fixed points

$$p_a=(a,a,b,b),\qquad b=\tfrac12-a,\qquad 0<a<\tfrac12. \tag{2}$$

For completeness, substitution into (1) gives

$$K(p_a)=\begin{pmatrix}a^2&2a-a^2&b^2+b&b^2+2ab\\2a-a^2&a^2&b^2+2ab&b^2+b\\a^2+a&a^2+2ab&b^2&2b-b^2\\a^2+2ab&a^2+a&2b-b^2&b^2\end{pmatrix}.$$

Multiplication gives $p_a K(p_a) = p_a$, using $a + b = 1/2$; uniqueness therefore gives $F(p_a) = p_a$.

Next we compute the derivative at $p_* = (1/4, 1/4, 1/4, 1/4)$. Use the following basis of the tangent space of the simplex:

$$e_0 = (1, 1, -1, -1), \quad e_1 = (1, -1, 1, -1), \quad e_2 = (1, -1, -1, 1).$$

At $p_*$, formula (1) yields

$$K_* = \frac{1}{16}\begin{pmatrix} 1 & 7 & 5 & 3 \\ 7 & 1 & 3 & 5 \\ 5 & 3 & 1 & 7 \\ 3 & 5 & 7 & 1 \end{pmatrix}, \qquad e_i K_* = \lambda_i e_i, \quad (\lambda_0, \lambda_1, \lambda_2) = (0, -1/4, -1/2).$$

Differentiating (1) also gives $p_* DK(p_*)[e_i] = e_i$. One way to check this last identity for any zero-sum vector $v$ is to note that every column of $R$ has ranks $1, 2, 3, 4$, and for every $z \neq y$ there are exactly two rows $x$ with $R_{xz} < R_{xy}$. Thus

$$(p_* DK(p_*)[v])_y = \frac{1}{8}(1 + 2 + 3 + 4)v_y + \frac{1}{8}\sum_{z \neq y} 2v_z = v_y.$$

Differentiating $F(p)K(p) = F(p)$ now shows that $w = DF(p_*)[e_i]$ satisfies $w(I - K_*) = e_i$ and $\sum_y w_y = 0$. Consequently

$$DF(p_*)[e_0] = e_0, \qquad DF(p_*)[e_1] = \frac{4}{5}e_1, \qquad DF(p_*)[e_2] = \frac{2}{3}e_2. \tag{3}$$

We finish with an explicit local convergence argument, so that the initial distributions need not lie in a symmetry subspace. In coordinates $p = p_* + te_0 + ue_1 + ve_2$, write $F$ as

$$(t, z) \longmapsto (T(t, z), G(t, z)), \qquad z = (u, v).$$

Equation (2) says $T(t, 0) = t$ and $G(t, 0) = 0$. By (3), $D_z G(0, 0) = \mathrm{diag}(4/5, 2/3)$. By continuity, choose $0 < \delta < 1/12$ so small that on the closed box $|t| \leq \delta$, $\|z\|_\infty \leq \delta$,

$$\|D_z G(t, z)\|_\infty \leq q := 9/10.$$

This box lies inside the simplex. Choose a finite $L \geq 1$ bounding the operator norm of $D_z T$ there. The mean-value inequality, applied along the segment from $(t, 0)$ to $(t, z)$, gives

$$\|G(t, z)\|_\infty \leq q\|z\|_\infty, \qquad |T(t, z) - t| \leq L\|z\|_\infty. \tag{4}$$

Start anywhere in the open set

$$|t_0| < \delta/2, \qquad \|z_0\|_\infty < \delta/(20L).$$

Induction using (4) proves that all iterates stay in the box:

$$\|z_n\|_\infty \leq q^n \|z_0\|_\infty, \qquad |t_n| \leq |t_0| + L\|z_0\|_\infty \sum_{m=0}^{n-1} q^m < \delta.$$

The increments $|t_{n+1} - t_n|$ are summable, hence $t_n \to t_\infty$ with $|t_\infty| < \delta$, while $z_n \to 0$. Therefore the iterates converge to $p_* + t_\infty e_0$, with all four coordinates positive. The displayed open set has positive three-dimensional Lebesgue measure. Its limits are neither point masses nor uniform two-point measures, contradicting Conjecture 5. □

The mechanism is attraction to a line of fixed points. An individual point of that line has a neutral tangential direction, while both transverse directions contract. The metric has distinct distances; its rank symmetries do not require the initial distribution to have any symmetry. This resolves the conjecture for arbitrary finite metrics by a counterexample and makes no assertion about the separate $\pi_{1,2}$ conjecture for planar Euclidean spaces in [1].

**Task 2. Determine the complete basin.** Characterize the initial distributions converging to the interior fixed line for this rank matrix, and determine the limiting parameter $a$ from the initial distribution.

**Task 3. Classify attracting fixed families.** Find conditions on a feasible rank matrix ensuring that $\pi_{2,2}$ has a positive-dimensional fixed family attracting a set of positive volume.

**Task 4. Analyze other sample ranks.** For this four-point metric, determine which pairs $(j, k)$ admit attracting limits supported on more than two points.

**Task 5. Identify geometric restrictions.** Determine which geometric classes of finite metrics exclude this mechanism, and formulate a corrected almost-everywhere convergence statement for them.

## References


[1] D. J. Aldous, M. Cruz and S. Feng, *Markov chains and mappings of distributions on compact spaces II: Numerics and Conjectures*, arXiv:2403.18153v1 (2024), Conjecture 5, pp. 14–15. https://arxiv.org/abs/2403.18153.

[2] D. J. Aldous and S. Feng, *Markov Chains and Mappings of Distributions on Compact Spaces*, ALEA **21** (2024), 1407–1416. https://doi.org/10.30757/ALEA.v21-53.